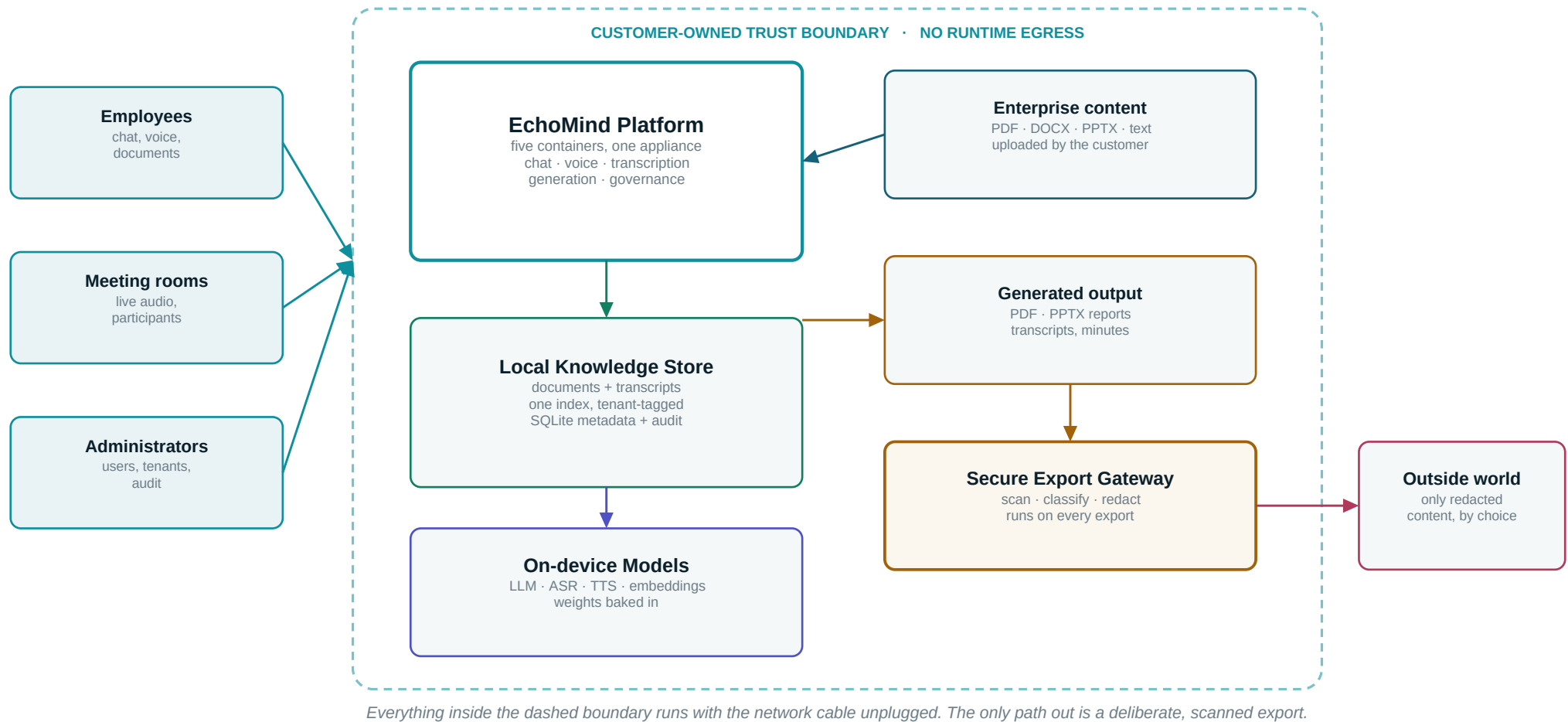


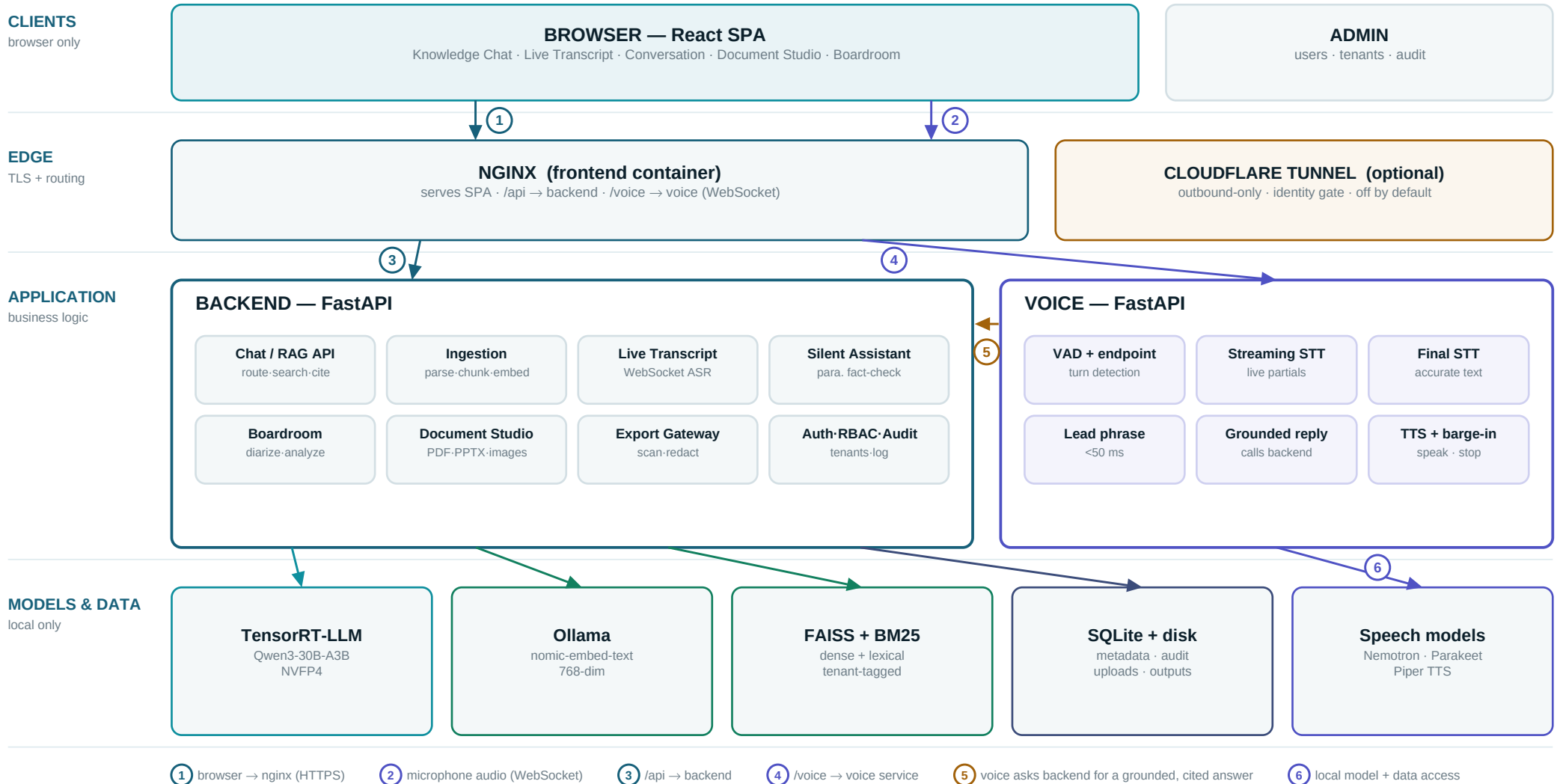
System Context

Who and what touches EchoMind, and where the trust boundary sits.



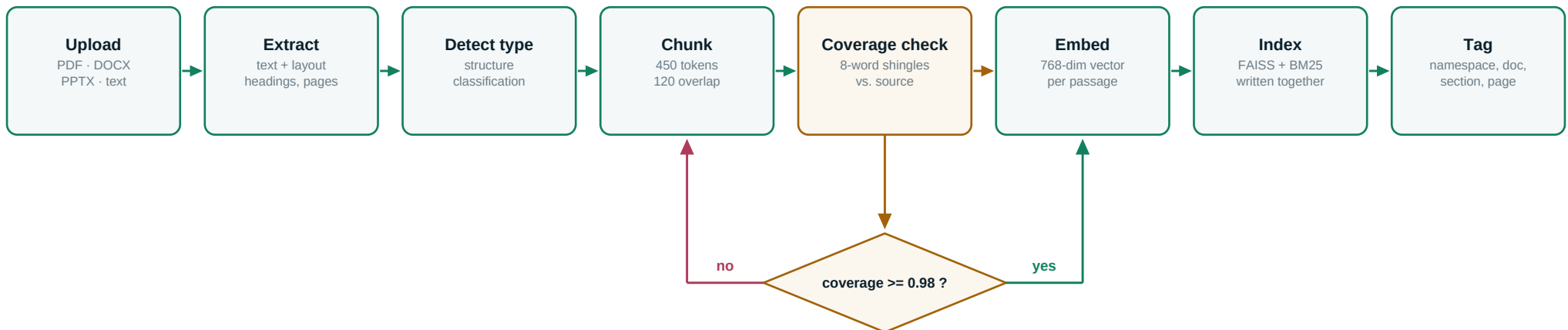
Runtime Architecture

The five containers, their responsibilities, and every internal call path.



Flow A — Document Ingestion

From an uploaded file to a searchable, tenant-tagged passage. Runs once per document.



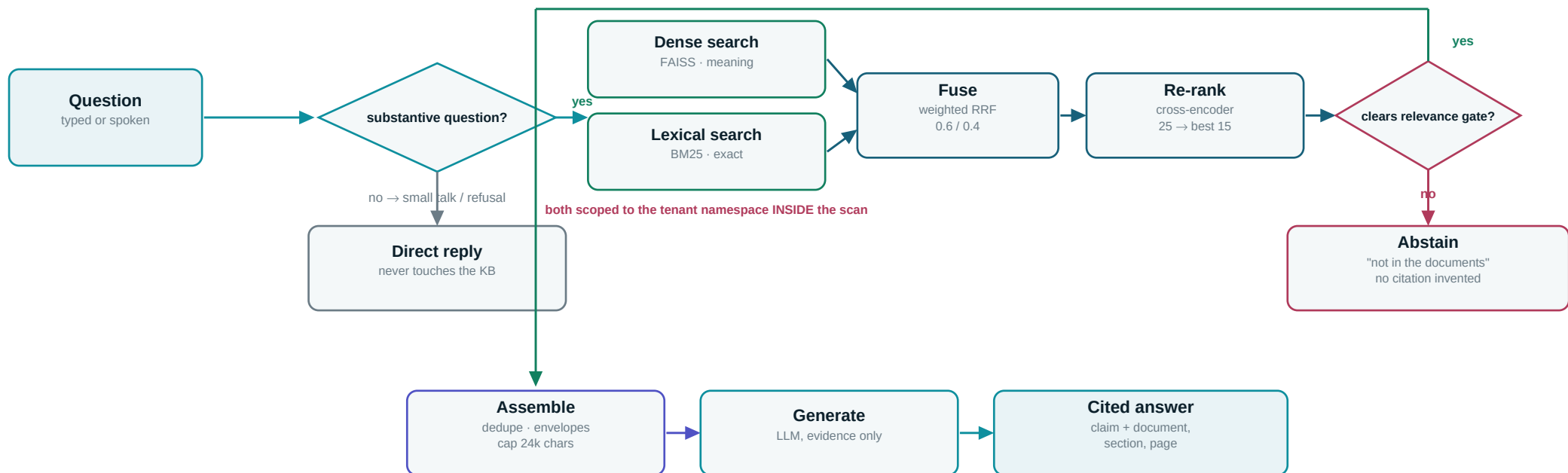
Below 0.98 the structured result is discarded and the proven flat splitter runs instead — content loss is never silent.

What each stage guarantees

Extract	Headings, sections and page numbers survive, so a citation can point at a real place in the file.
Chunk	450-token windows with 120-token overlap; a sentence spanning a boundary is never lost.
Coverage check	Measures how much of the original text the chunks actually cover; below 98% the structured path is discarded.
Embed / Index	Filed twice — vector index for meaning, lexical index for exact terms. Both are needed.
Tag	Namespace written at index time, so nothing downstream has to remember to filter.

Flow B — Question to Cited Answer

The governed retrieval path, including the abstention branch.

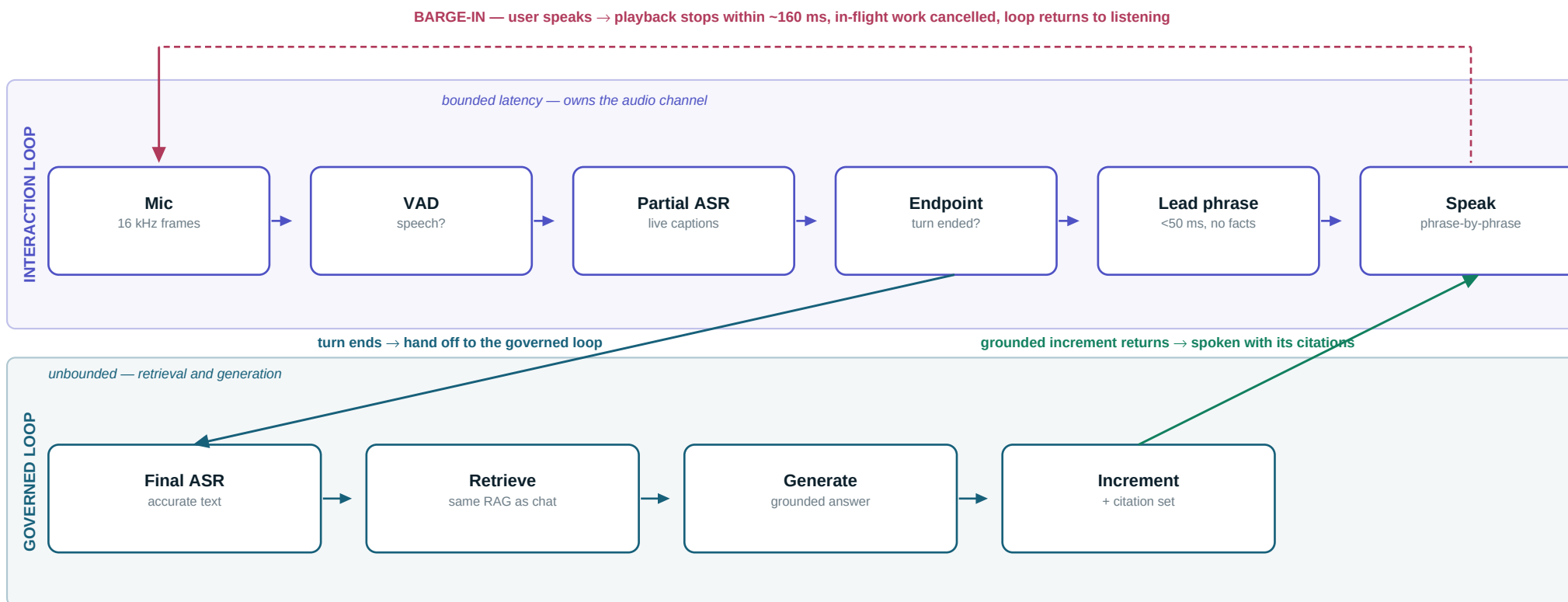


Two structural properties worth noting:

- The tenant predicate runs inside the candidate scan on all five retrieval paths — not as a filter on ranked results — so another tenant's passage never enters the ranking at all.
- The relevance gate has a real 'no' branch. If nothing clears it the context is dropped and the assistant abstains, rather than being handed weak evidence and asked to do its best.

Flow C — The Dual-Loop Speech Cycle

How the assistant stays responsive without asserting anything it has not retrieved.



Why two loops

A single loop must stay silent until the grounded answer is ready. That silence is what makes voice assistants feel broken.

The safety rule

The interaction loop may acknowledge, but may never assert a fact retrieval has not returned. Lead phrases carry no content.

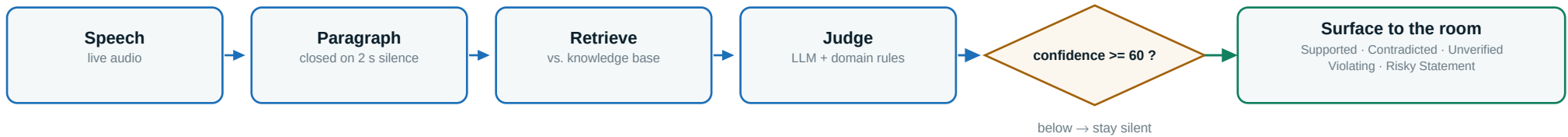
Measured effect

636 ms to first audio with the dual loop vs 866 ms single-loop. Removing the safety rule produced ungrounded claims on 10 of 10 turns.

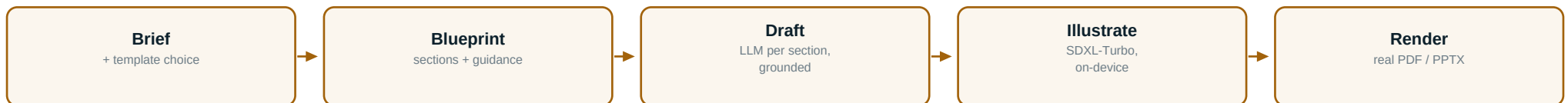
Flow D — Live Assurance, Generation and Egress

Three supporting flows: the Silent Assistant, Document Studio, and the export gate.

SILENT ASSISTANT — runs alongside every meeting



DOCUMENT STUDIO — brief to finished file



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SECURE EXPORT GATEWAY — the only way content leaves

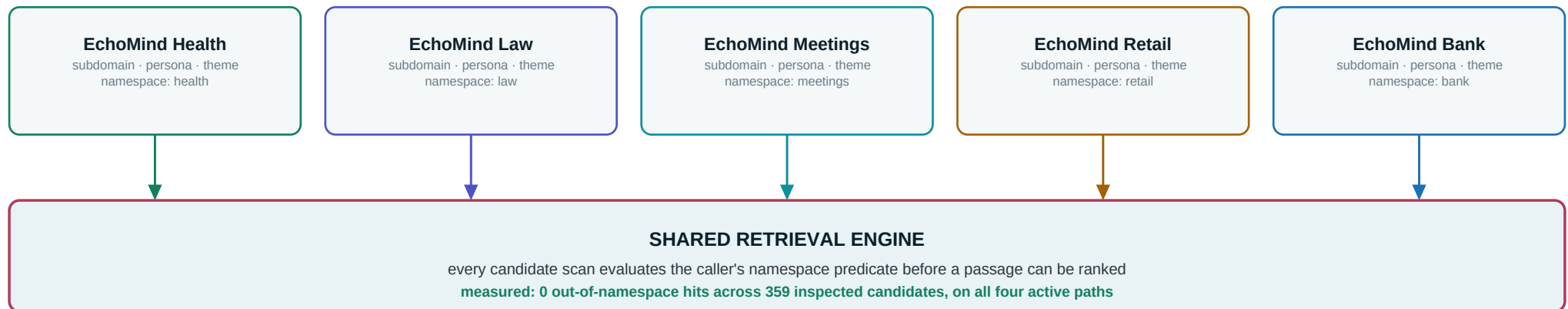


Regex + Luhn only — no GPU, no network — so it runs on every export, including fully air-gapped deployments.

Multi-Tenant Isolation and Deployment

How one appliance serves several products without their data meeting.

TENANT ISOLATION — enforced inside the query, not after it



DEPLOYMENT MODELS



OPERATIONAL PROPERTIES

- One command starts all five containers; public access is a separate profile, off by default.
- Health checks detect fatal GPU faults and restart with a clean CUDA context rather than degrading silently.
- A 52-question golden suite gates every release; a drop in pass rate blocks the change.
- Model weights ship inside the images — no download at startup, no registry dependency.
- Knowledge base, model cache and embedding store live in named volumes and survive rebuilds.